JULY 29, 2026 ACCELERATOR & HBM MEMORY MODEL

Rubin Ultra HBM to 192GB // Mutli-Vertical Note: Accelerator and HBM Model and Memory Model

2 MINUTES

By Myron Xie

According to industry chatter, Rubin "Ultra" is being previewed to key customers with lower specifications than previously expected. The mainline Rubin Ultra SKU will now have the same peak theoretical FLOPs, remain on HBM4 but is de-specced to 8-Hi 192GB, which is lower than 288GB 12-Hi Rubin. There is slightly more memory bandwidth and chip level power is at ~1,800W, which is the same power as the initial Rubin shipments but lower than the 2,300W version with 2-piece lid. This means the only upgrade for

Rubin Ultra on the current roadmap is NVL576 scale up world size. One of the planned system form factors to get to NVL576 is 8x 72 GPU racks interconnected, with NPO for switch-to-switch inter-rack connections. As always, Nvidia has multiple development tracks, but we believe this form factor is seen as the most promising candidate.

The Rubin Ultra logic silicon is being designed for higher power and we believe Nvidia may still offer a version that is 2,600-2,800W, but the 1,800W version is the mainline offering. There may be an even lower power 1,200W SKU that would be geared towards lower compute workloads such as token decode. The 2,600W+ SKU will be offered as the Max-P SKU, and the 1,800W SKU will be offered as the Max-Q SKU. The similar setup happened with Rubin, where the Max-P SKU can operate at higher TDP and higher absolute flops, while the Max-Q SKU operates at a lower TDP but offers optimized FLOPs per watt.

Rubin Family Spec Comparison

	Rubin	Rubin "Ultra"	Rubin Ultra vs Rubin
NVFP4 PFLOPs	35	35	Same
Chip TDP (W)	1,800 or 2,300	1,800 or 2,600	Mainline SKU the same
HBM type	HBM4 12-Hi	HBM4 8-Hi	
HBM capacity (GB)	288	192	Lower
HBM bandwidth (TB/s)	20*	21	Slight increase
Scale-up world size (# GPUs)	72	576	Significantly larger

**Rubin is marketed as up to 22TB/s memory bandwidth but will mostly ship with 20TB/s which is 9.7Gbps pin speed*

Source: SemiAnalysis

At the package level, this continues the trend of specifications downgrades from the original 4-die 1TB HBM4E 16-Hi version that was first previewed at GTC 2025. First the HBM4E spec was downgraded to HBM4E 12-Hi. Subsequently, the 4-die MCM was also canceled so that Rubin Ultra remains the same 2 die package as Rubin. While some of these downgrades have been due to manufacturing challenges, this Rubin Ultra is even lower spec than Rubin, so these are voluntary choices.

Rubin Family Roadmap Progression

		Rubin MaxP	Rubin MaxQ	Rubin Ultra Intial 2025 GTC Acnnouncement	Rubin Ultra Cut HBM To 12-Hi	2 Die Rubin Ultra HBM4E 12-Hi	Current Mainline: Rubin Ultra 1,800W (Max-Q)
Number of Compute Die		2	2	4	4	2	2
HBM generation		HBM4	HBM4	HBM4E	HBM4E	HBM4E	HBM4
Number of HBM stacks		12	12	16	12	12	8
HBM capacity (GB) at given stack height, mainline SKU highlighted	8	192	192	512	512	256	192
	12	288	288	768	768	384	288
	16	384	384	1,024	1,024	512	384
TDP (W)		2,300	1,800	3,600	3,600	2,800	1,800

Source: SemiAnalysis

We believe that the rationale for Nvidia adjusting the roadmap in such a way is based on the intersection of memory and DC supply challenges. On the memory side this is the same rationale as the initial Rubin Ultra reduction from 16-hi to 12-hi, and reducing SOCAMM per Vera. It is rationing supply to make the most of the limited HBM on offer from suppliers and helps balance relatively less tight TSMC front end supply with relatively more tight HBM supply. This also offers a significant BOM reduction that offsets the HBM price hikes coming next year. This is even though Nvidia has secured a significant $/GB discount on HBM from SK Hynix relative to other customers in exchange for providing GPU allocation for SK Group's AI Datacenter initiative.

On the power side, with our AI Datacenter model showing continued power deficits, Nvidia is reducing the power spec to make it easier for customers to deploy. Although the Rubin Ultra tape out (codenamed GR150) is designed to handle much higher power than the original GR100 Rubin, it is likely that most customers won't be able to make the most use of the additional power budget. Under a power constrained world, the 1,800W Max-Q SKU is the most optimized option, as it is the most efficient in terms of FLOPs per Watt. Said another way, even if Nvidia offered the higher power 2,600W Rubin Ultra as the mainline SKU, most customers would not choose to operate near max power, therefore Nvidia can reduce power delivery BOM and also harvest higher wafer yields.

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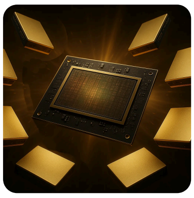

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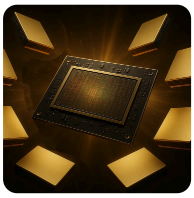

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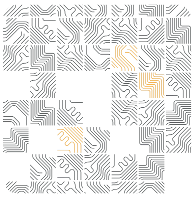

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